

20 jobs

Yamada and Nakano 20x20 instance (Table 4, instance 4)

20 machines

|    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 16 | 34 | 17 | 38 | 0  | 21 | 6  | 15 | 15 | 42 | 8  | 17 | 7  | 41 | 18 | 10 | 10 | 26 | 11 | 24 | 1  | 31 | 19 | 25 | 14 | 31 | 13 | 33 | 4  | 35 | 9  | 30 | 3  | 16 | 12 | 16 | 5  | 30 | 2  | 13 |
| 5  | 41 | 11 | 33 | 6  | 15 | 16 | 38 | 0  | 40 | 14 | 38 | 3  | 37 | 1  | 20 | 13 | 22 | 4  | 34 | 7  | 16 | 17 | 39 | 9  | 15 | 2  | 19 | 10 | 36 | 12 | 39 | 18 | 26 | 8  | 19 | 15 | 39 | 19 | 34 |
| 17 | 34 | 1  | 12 | 16 | 10 | 7  | 47 | 13 | 28 | 15 | 27 | 0  | 19 | 6  | 34 | 19 | 33 | 12 | 40 | 9  | 37 | 14 | 24 | 8  | 15 | 10 | 34 | 2  | 44 | 3  | 37 | 18 | 22 | 11 | 31 | 4  | 39 | 5  | 26 |
| 5  | 48 | 7  | 46 | 16 | 47 | 10 | 45 | 14 | 15 | 8  | 25 | 0  | 34 | 3  | 24 | 12 | 35 | 18 | 15 | 2  | 48 | 13 | 19 | 11 | 10 | 1  | 48 | 17 | 16 | 15 | 28 | 4  | 18 | 6  | 17 | 9  | 44 | 19 | 41 |
| 12 | 47 | 3  | 23 | 9  | 48 | 16 | 45 | 14 | 39 | 6  | 42 | 8  | 32 | 15 | 11 | 13 | 16 | 5  | 14 | 11 | 19 | 1  | 46 | 19 | 10 | 10 | 17 | 7  | 41 | 2  | 47 | 17 | 32 | 4  | 17 | 0  | 21 | 18 | 17 |
| 18 | 14 | 16 | 20 | 1  | 18 | 12 | 14 | 13 | 10 | 6  | 16 | 5  | 24 | 4  | 18 | 0  | 24 | 11 | 18 | 15 | 42 | 19 | 13 | 3  | 23 | 14 | 40 | 9  | 48 | 8  | 12 | 2  | 24 | 10 | 23 | 7  | 45 | 17 | 30 |
| 0  | 27 | 12 | 15 | 4  | 26 | 13 | 19 | 17 | 14 | 5  | 49 | 7  | 16 | 18 | 28 | 16 | 16 | 8  | 20 | 9  | 36 | 2  | 21 | 14 | 30 | 3  | 36 | 1  | 17 | 15 | 22 | 6  | 43 | 11 | 32 | 10 | 23 | 19 | 17 |
| 0  | 32 | 16 | 15 | 17 | 12 | 7  | 46 | 3  | 37 | 18 | 43 | 11 | 40 | 13 | 43 | 9  | 48 | 4  | 36 | 15 | 24 | 8  | 25 | 1  | 33 | 14 | 32 | 5  | 26 | 6  | 37 | 12 | 24 | 10 | 24 | 2  | 15 | 19 | 22 |
| 10 | 34 | 6  | 33 | 15 | 25 | 8  | 46 | 0  | 20 | 18 | 33 | 4  | 19 | 13 | 45 | 2  | 47 | 1  | 32 | 3  | 12 | 11 | 29 | 16 | 29 | 5  | 46 | 12 | 17 | 7  | 48 | 14 | 39 | 17 | 40 | 19 | 41 | 9  | 37 |
| 13 | 26 | 3  | 47 | 5  | 44 | 6  | 49 | 1  | 22 | 17 | 12 | 10 | 28 | 19 | 36 | 9  | 27 | 4  | 25 | 14 | 48 | 7  | 11 | 16 | 49 | 12 | 24 | 11 | 48 | 2  | 19 | 0  | 47 | 18 | 49 | 8  | 46 | 15 | 36 |
| 13 | 23 | 18 | 48 | 14 | 15 | 0  | 42 | 3  | 36 | 8  | 15 | 6  | 32 | 10 | 18 | 1  | 45 | 15 | 23 | 11 | 45 | 2  | 13 | 17 | 21 | 12 | 32 | 7  | 44 | 5  | 25 | 19 | 34 | 16 | 22 | 9  | 11 | 4  | 43 |
| 17 | 37 | 7  | 49 | 15 | 45 | 2  | 28 | 9  | 15 | 8  | 35 | 12 | 29 | 13 | 44 | 1  | 26 | 4  | 25 | 5  | 30 | 3  | 39 | 0  | 15 | 14 | 28 | 18 | 23 | 6  | 42 | 11 | 33 | 16 | 45 | 10 | 10 | 19 | 20 |
| 0  | 10 | 6  | 37 | 3  | 15 | 13 | 13 | 10 | 11 | 2  | 49 | 1  | 28 | 14 | 28 | 15 | 13 | 8  | 29 | 12 | 21 | 16 | 32 | 11 | 21 | 4  | 48 | 5  | 11 | 17 | 26 | 9  | 33 | 18 | 22 | 7  | 21 | 19 | 49 |
| 18 | 38 | 0  | 41 | 4  | 30 | 13 | 43 | 6  | 11 | 2  | 43 | 14 | 27 | 3  | 26 | 9  | 30 | 15 | 19 | 16 | 36 | 1  | 31 | 17 | 47 | 5  | 41 | 10 | 34 | 8  | 40 | 12 | 32 | 7  | 13 | 11 | 18 | 19 | 27 |
| 6  | 24 | 5  | 30 | 7  | 10 | 10 | 35 | 8  | 28 | 16 | 43 | 19 | 12 | 9  | 44 | 15 | 15 | 3  | 15 | 2  | 35 | 18 | 43 | 0  | 38 | 4  | 16 | 1  | 29 | 17 | 40 | 14 | 49 | 13 | 38 | 12 | 16 | 11 | 30 |
| 3  | 48 | 6  | 35 | 13 | 43 | 2  | 37 | 17 | 18 | 5  | 27 | 9  | 27 | 7  | 41 | 1  | 22 | 15 | 28 | 16 | 18 | 10 | 37 | 18 | 48 | 4  | 10 | 8  | 14 | 11 | 18 | 14 | 43 | 0  | 48 | 12 | 12 | 19 | 49 |
| 0  | 13 | 13 | 38 | 7  | 34 | 6  | 42 | 1  | 36 | 5  | 45 | 18 | 24 | 8  | 35 | 14 | 26 | 19 | 30 | 12 | 47 | 16 | 24 | 11 | 47 | 4  | 40 | 10 | 43 | 3  | 16 | 15 | 10 | 2  | 12 | 9  | 39 | 17 | 22 |
| 16 | 30 | 13 | 47 | 19 | 49 | 8  | 20 | 4  | 40 | 3  | 46 | 17 | 21 | 14 | 33 | 6  | 44 | 7  | 23 | 9  | 24 | 0  | 48 | 10 | 43 | 15 | 41 | 2  | 32 | 5  | 29 | 11 | 36 | 1  | 38 | 12 | 47 | 18 | 12 |
| 13 | 10 | 5  | 36 | 12 | 18 | 16 | 48 | 0  | 27 | 14 | 43 | 10 | 46 | 6  | 27 | 7  | 46 | 19 | 35 | 11 | 31 | 2  | 18 | 8  | 24 | 3  | 23 | 17 | 29 | 18 | 14 | 9  | 19 | 1  | 40 | 15 | 38 | 4  | 13 |
| 9  | 45 | 16 | 44 | 0  | 43 | 17 | 31 | 14 | 35 | 13 | 17 | 12 | 42 | 3  | 14 | 18 | 37 | 10 | 39 | 6  | 48 | 7  | 38 | 15 | 26 | 4  | 49 | 2  | 28 | 11 | 35 | 1  | 42 | 5  | 24 | 8  | 44 | 19 | 38 |